

Lab 01

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Reg No: 2024-BSE-052

Course Code: Cloud Computing

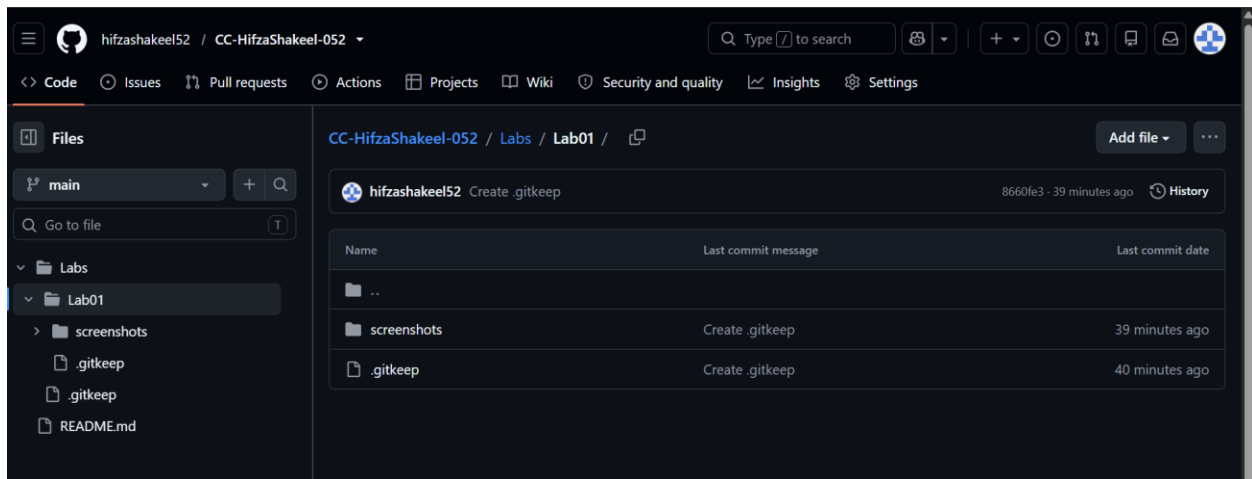
Section: BSE-5B

GitHub Username: hifzashakeel52

Lab Title: **GitHub Identity, Student Portal Registration, and Ubuntu Server Installation**

Directory Structure Verification

Before starting the lab tasks, the project workspace and target directory structure were created under CC/Labs/Lab01/screenshots/.



Created directory structure showing CC/Labs/Lab01/screenshots/.

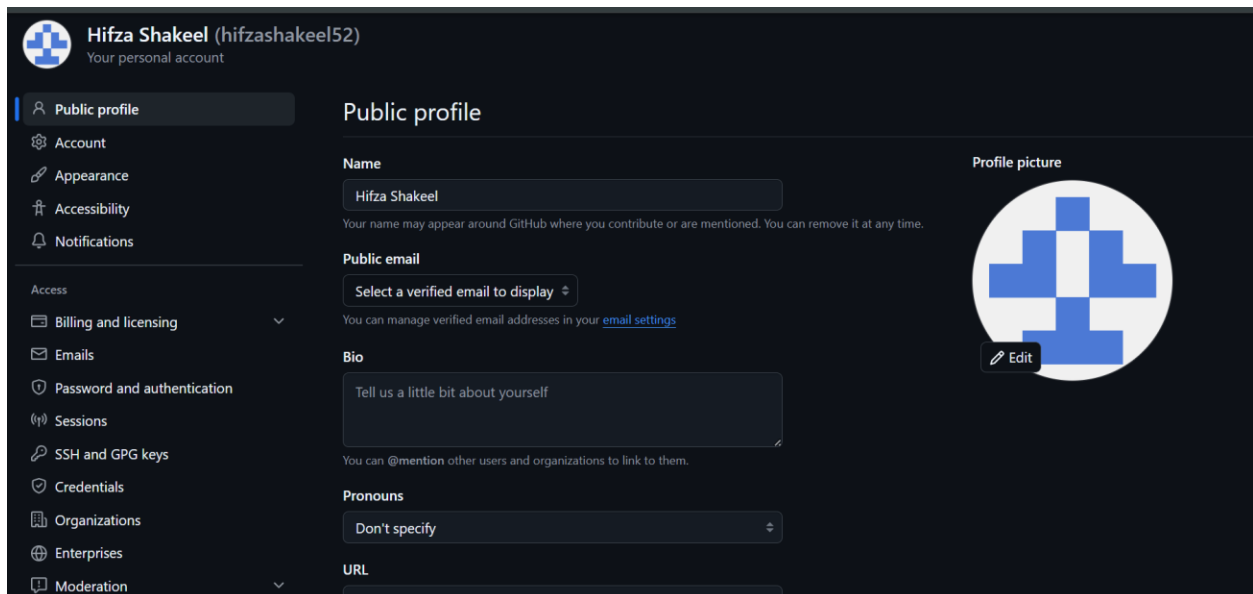
Task 1: Create and Configure Your GitHub Account

1. Created a personal GitHub account adhering to the professional username policy and actual identity.



GitHub profile showing the chosen username and profile overview.

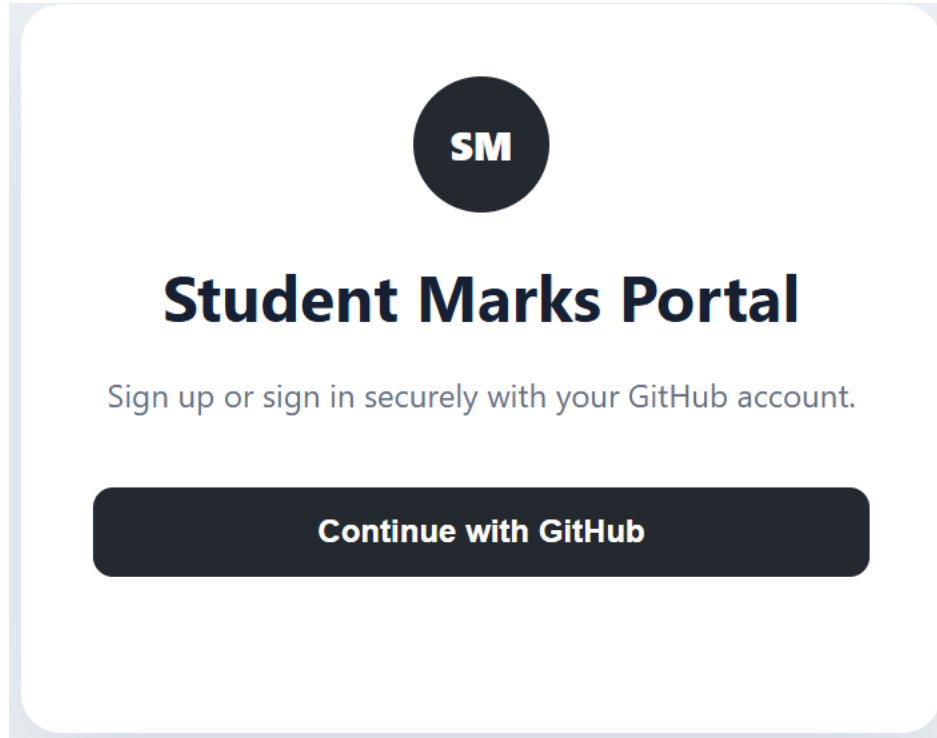
2. Configured the public profile with the complete actual name in the settings menu.



GitHub Public Profile settings showing the complete actual name field filled out.

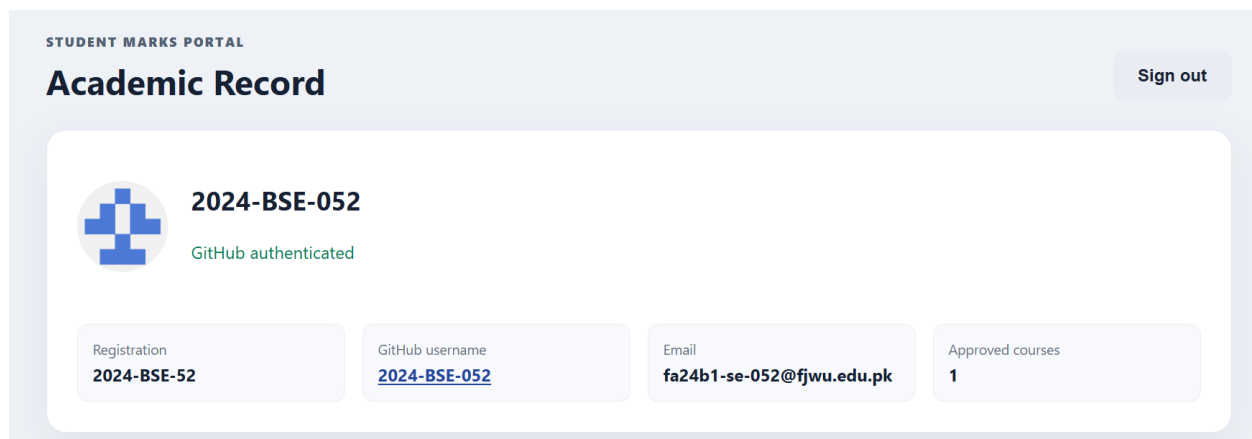
Task 2: Register in the Student Marks Portal

1. Navigated to the Student Marks Portal authentication page.



Student Marks Portal login page.

2. Authenticated using the GitHub account created in Task 1 and verified profile details inside the portal.



GitHub profile information reflected inside the Student Marks Portal.

3. Completed registration by entering the registration number, course code, and section, then submitted for approval.

+ Enroll in another course

APPROVED COURSES

Course Marks

Published

Cloud Computing (SE-665) Section B

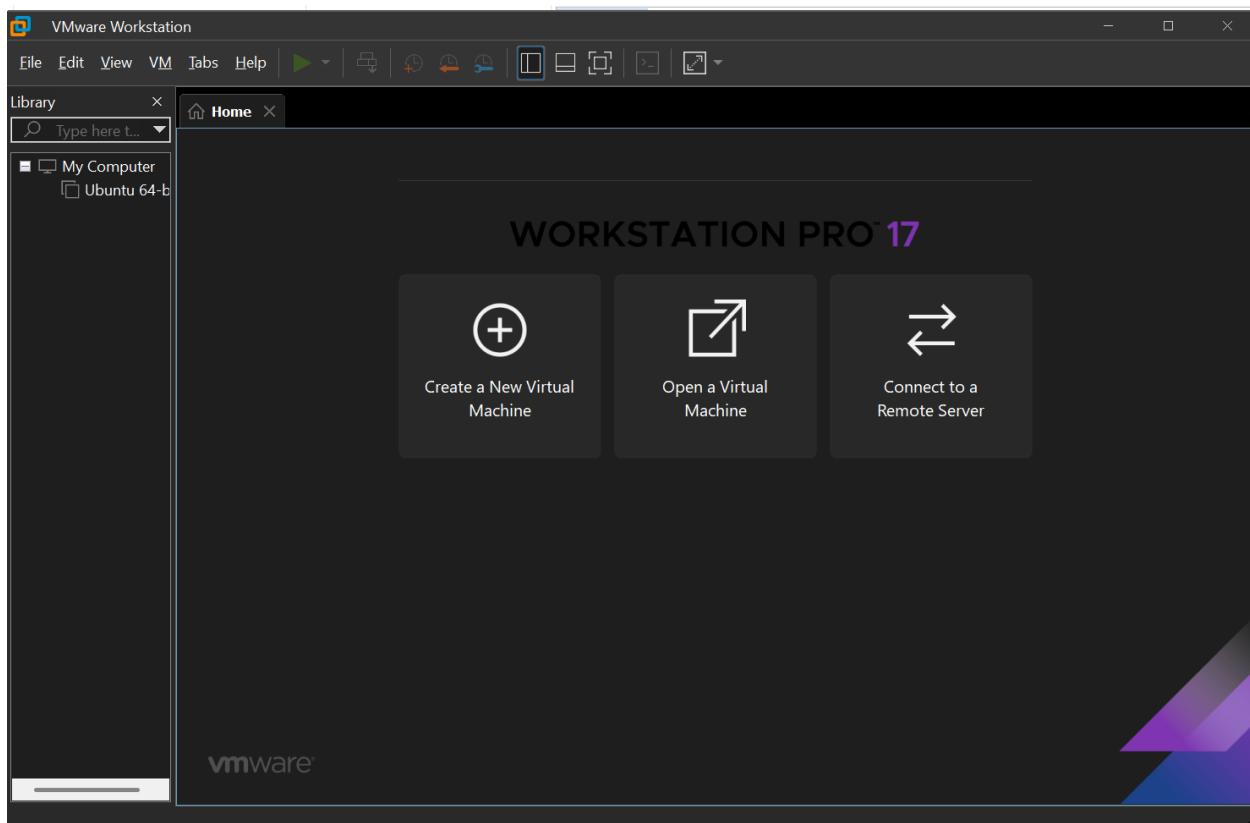
Class Participation

ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Class Participation	0	10	0.0%
Total	0	10	0.0%

Submitted enrollment screen showing registration details and pending approval status.

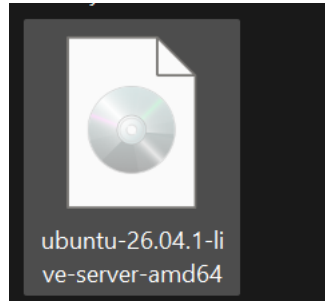
Task 3: Download VMware Workstation Pro and Ubuntu Server

1. Launched VMware Workstation Pro on the host Windows system.



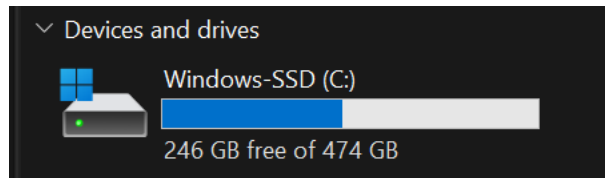
VMware Workstation Pro running on Windows.

2. Downloaded the latest Ubuntu Server LTS ISO file to local storage.



File explorer showing the downloaded Ubuntu Server LTS ISO file.

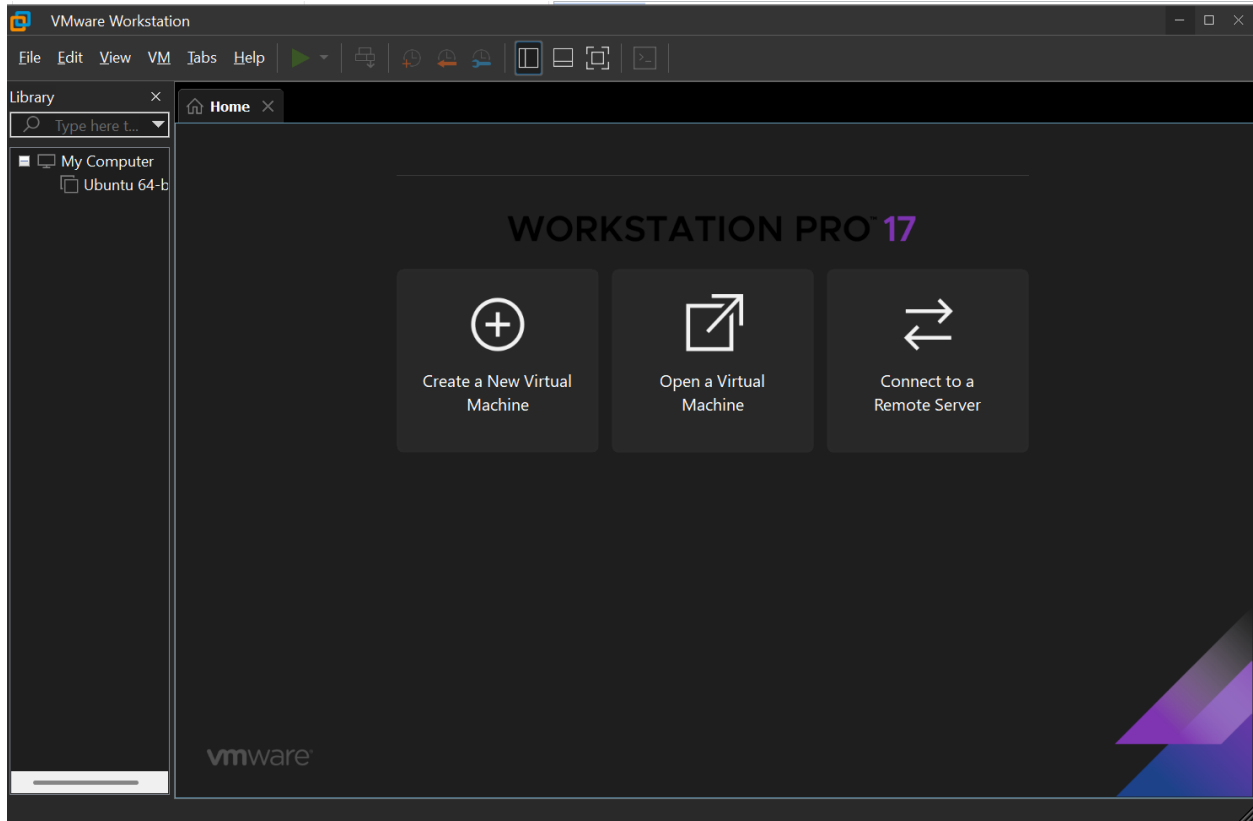
3. Checked and verified available host storage capacity to ensure at least 20 GB free space.



System drive properties demonstrating sufficient free storage space.

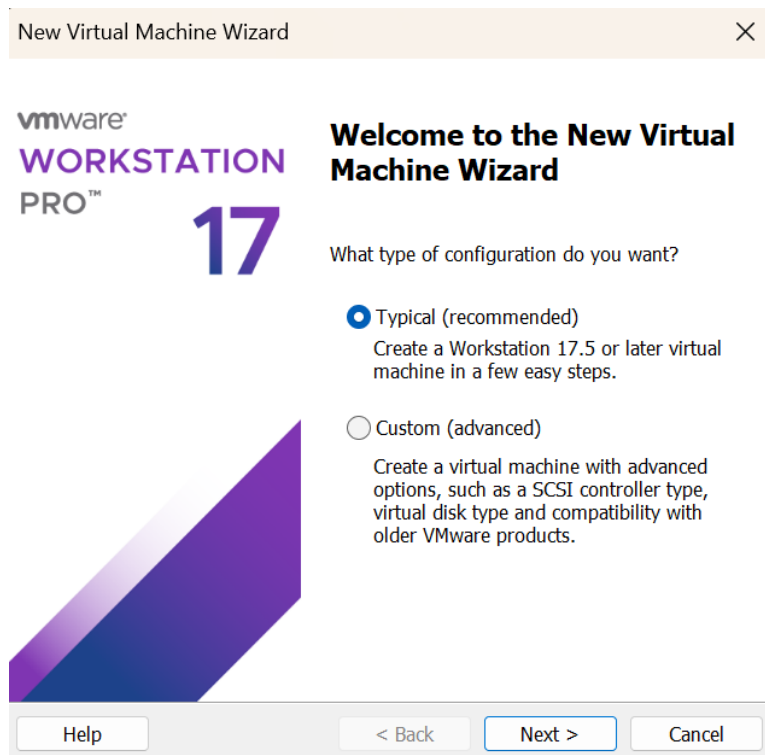
Task 4: Create the Ubuntu Server Virtual Machine

1. Opened the New Virtual Machine Wizard in VMware.



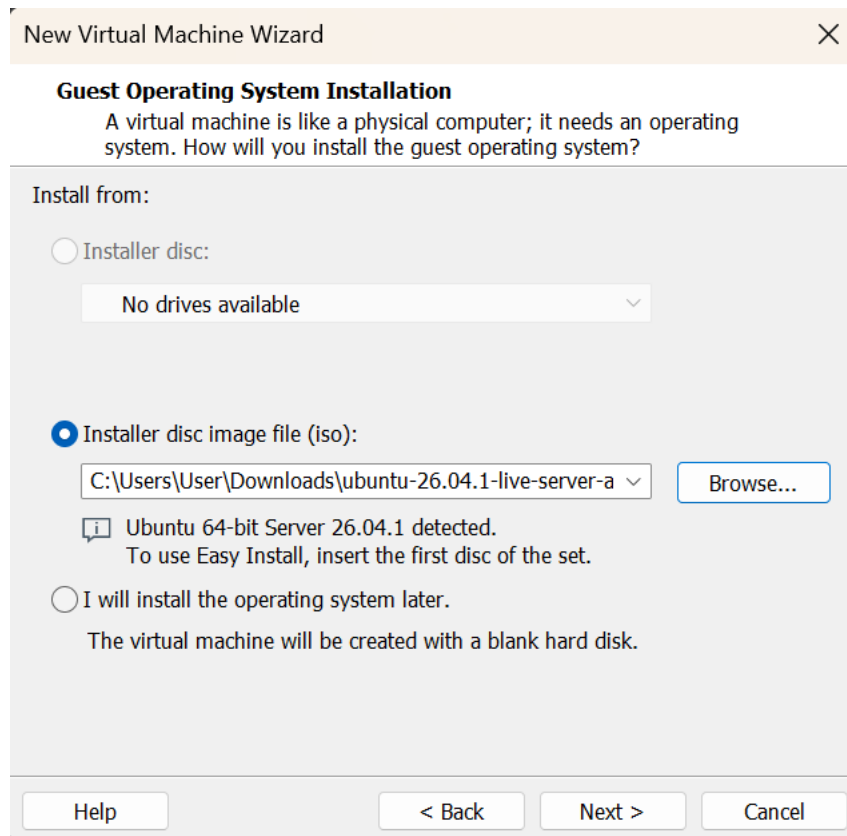
VMware New Virtual Machine Wizard startup screen.

2. Selected the Typical (recommended) configuration mode.



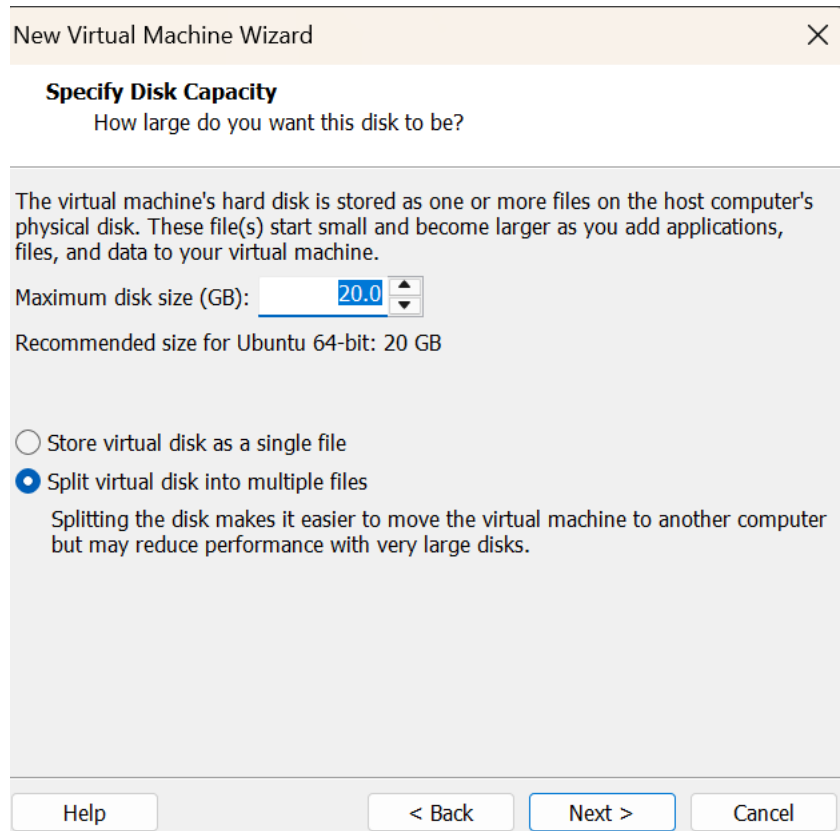
Selection of Typical virtual machine configuration type.

3. Attached the downloaded Ubuntu Server ISO file as the installer disc image.



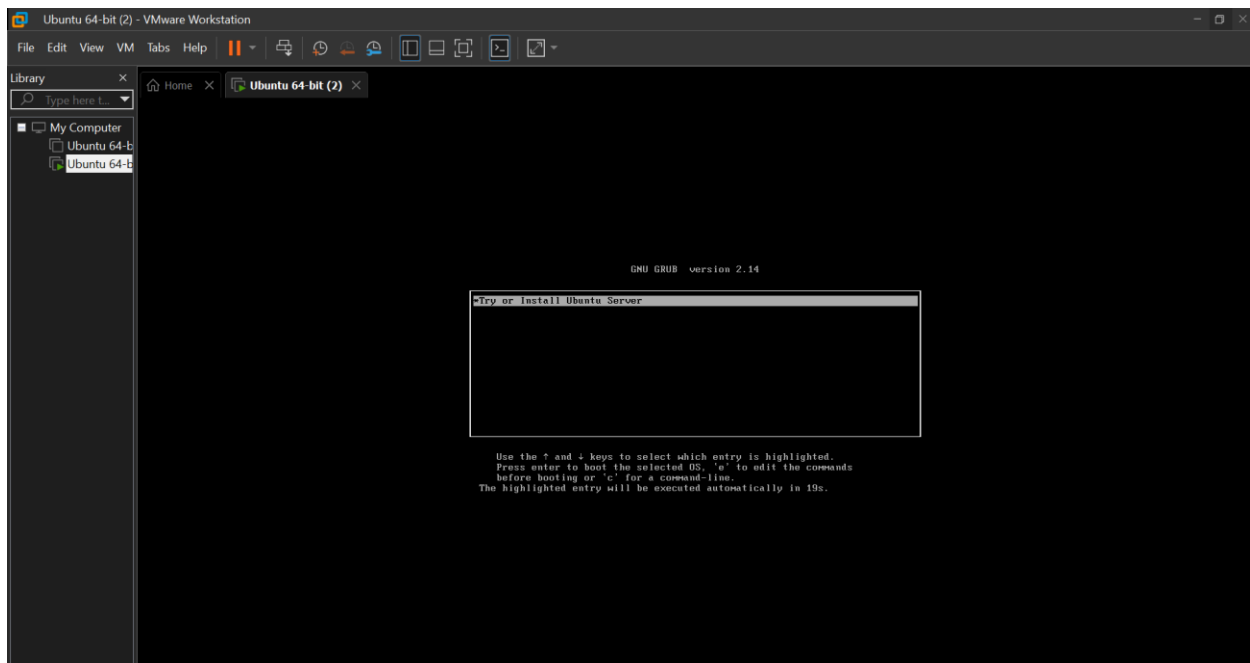
ISO path selection screen pointing to Ubuntu Server ISO.

4. Configured disk allocation (20 GB minimum) and finalized the wizard settings.

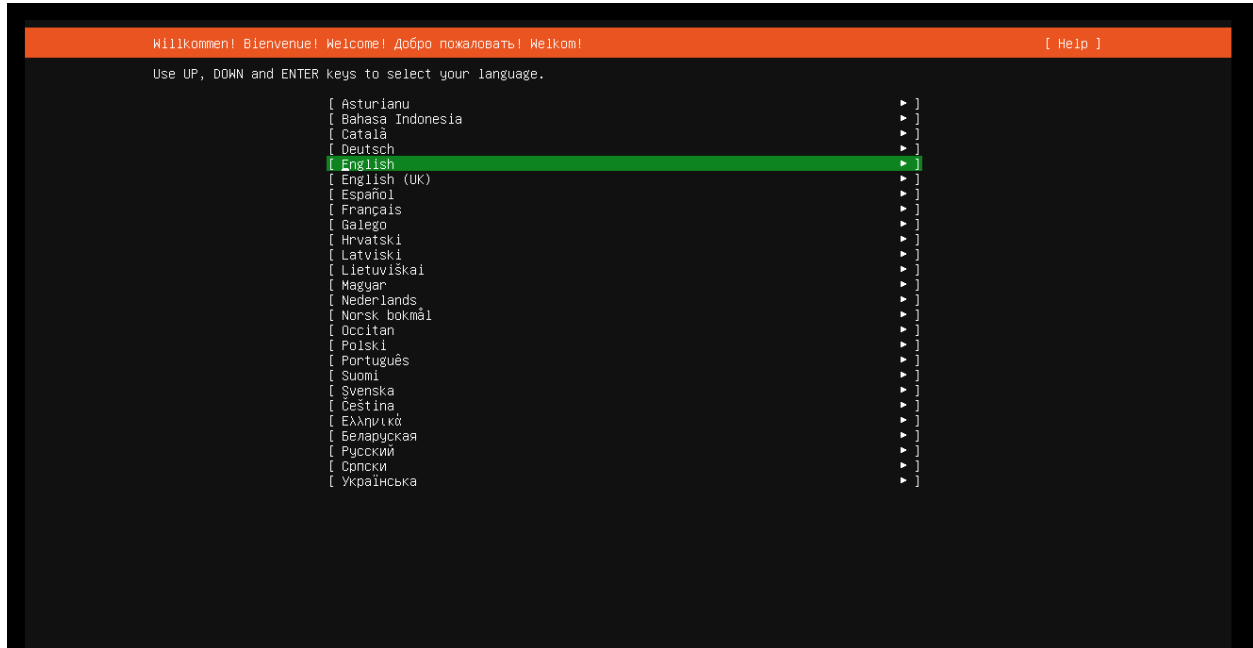


Final summary of virtual machine hardware configuration.

5. Powered on the virtual machine to initiate the boot process.



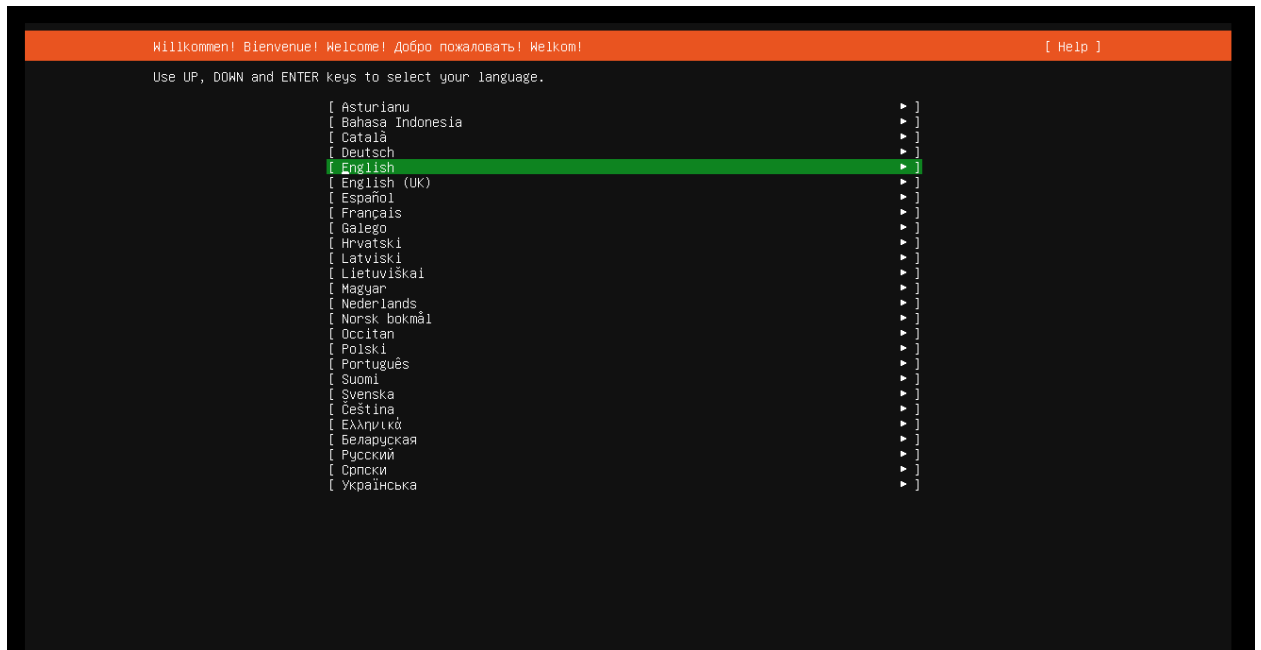
2. Selected the appropriate keyboard variant.



Keyboard variant selection step.

Task 5: Choose the Installation Language

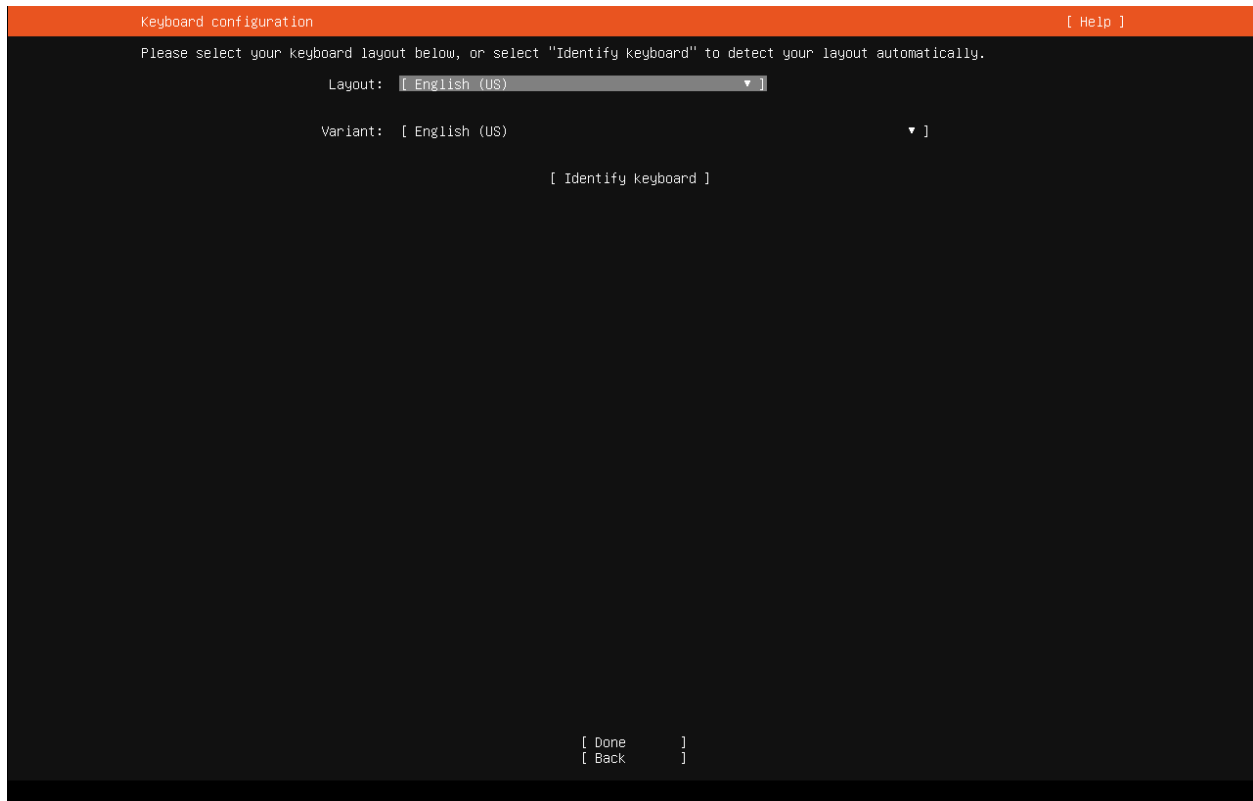
1. Selected the primary installation language using the keyboard navigation options.



Language selection interface during Ubuntu installation setup.

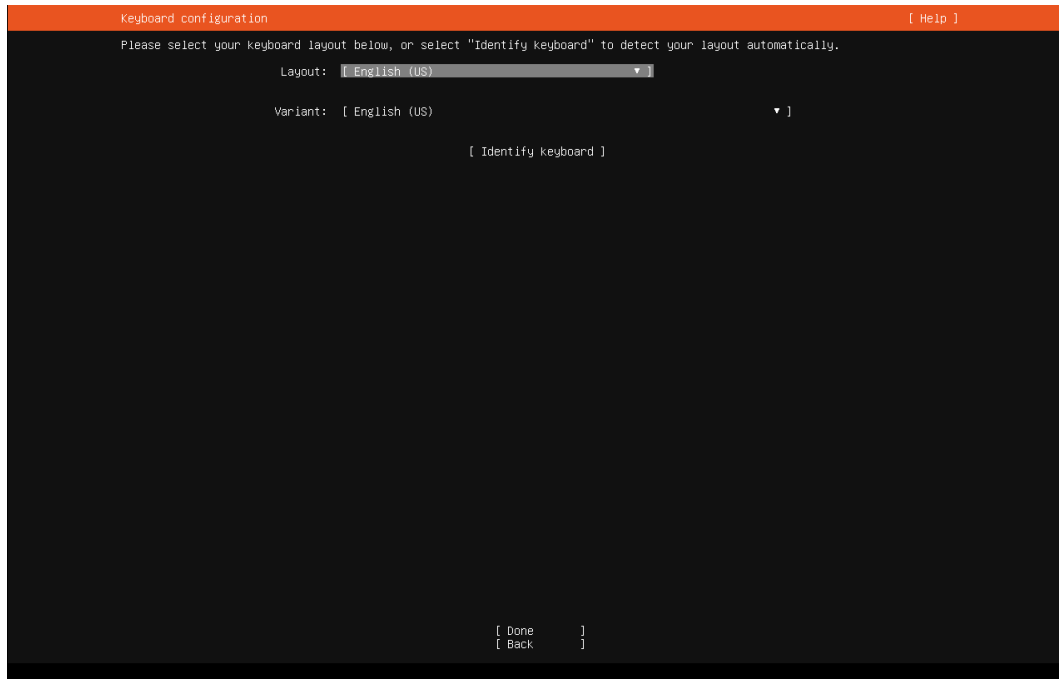
Task 6: Configure the Keyboard Layout

1. Selected the appropriate keyboard layout matching the hardware input setup.



Keyboard layout configuration screen.

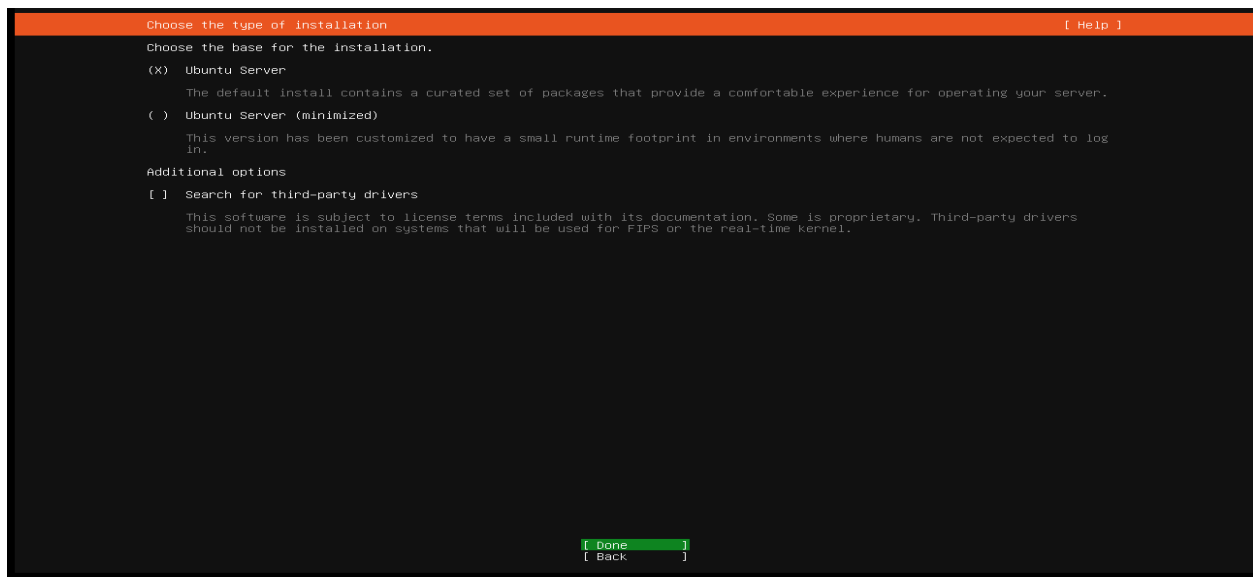
2. Selected the appropriate keyboard variant.



Keyboard variant selection step.

Task 7: Choose the Ubuntu Server Installation Type

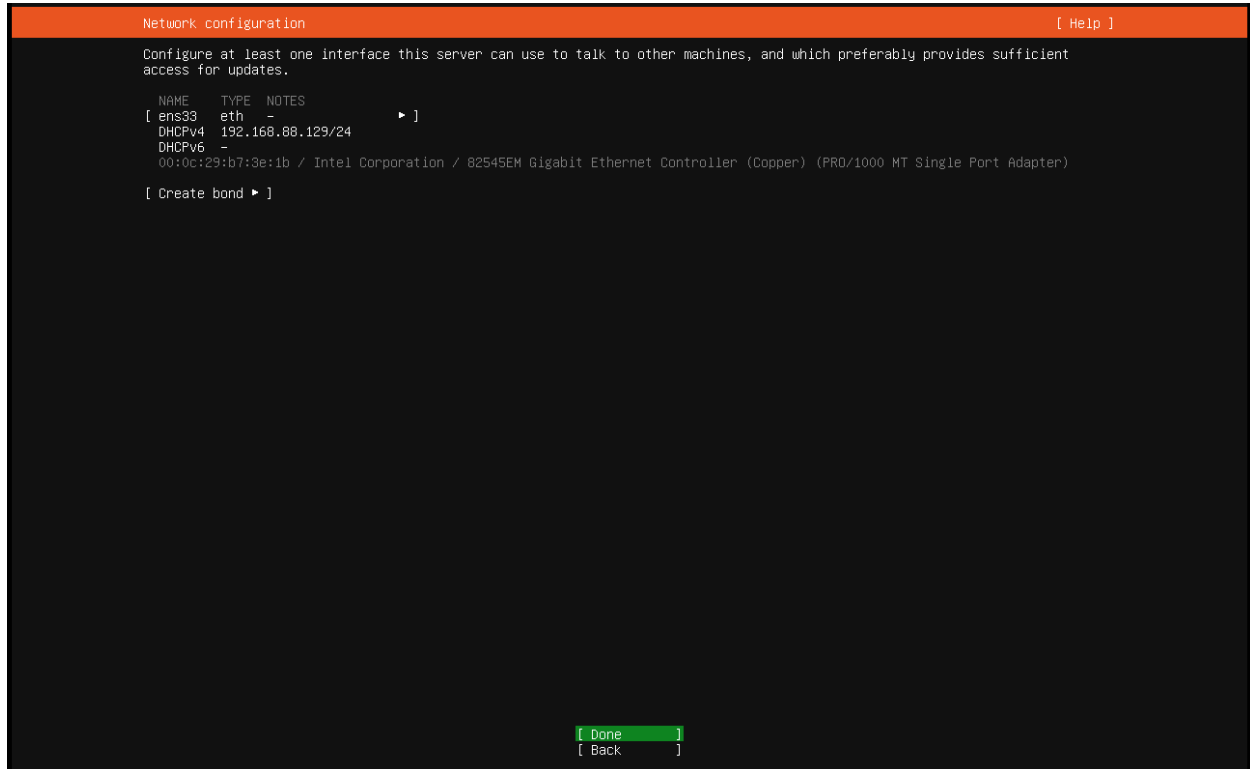
1. Selected the standard base installation type for Ubuntu Server.



Ubuntu Server installation mode selection.

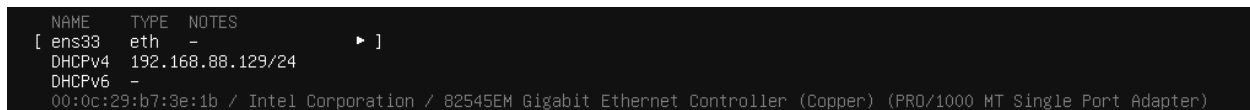
Task 8: Configure the Network

1. Verified the automatically detected network interface controller (NIC).



Installer detecting system network interfaces.

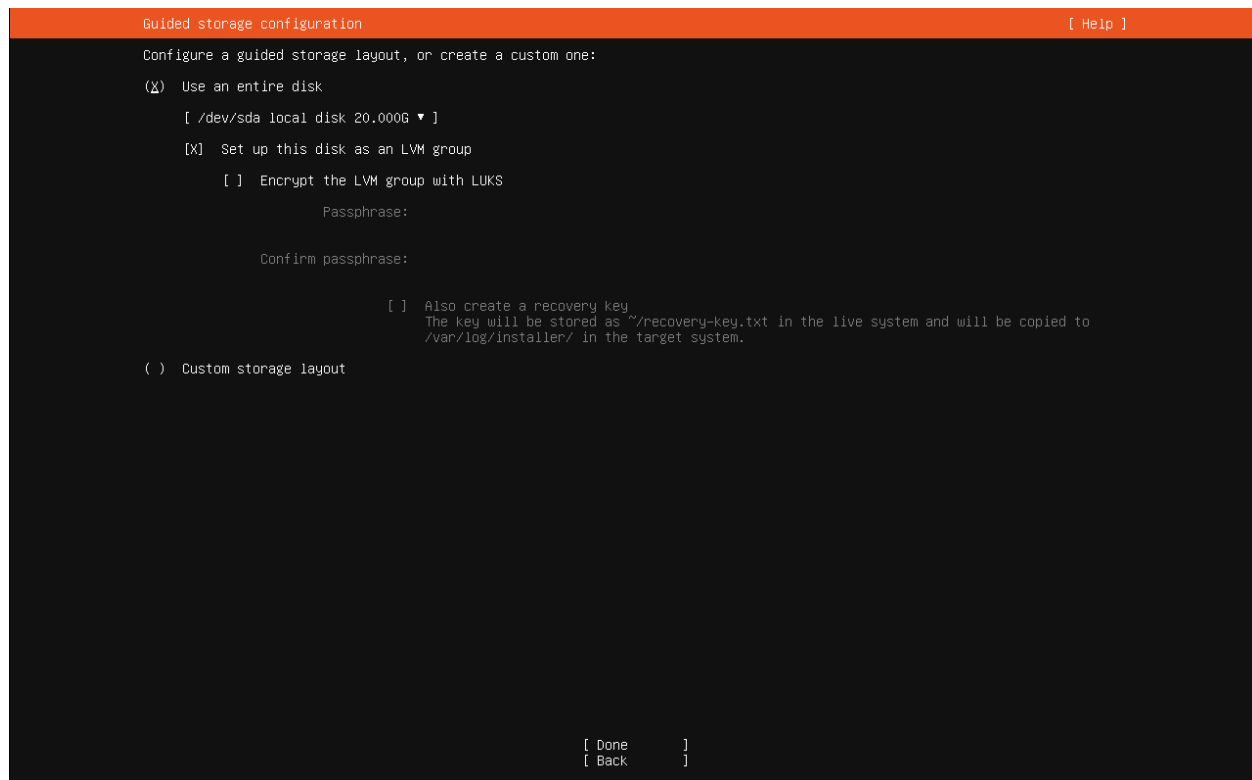
2. Confirmed automatic network configuration and IP assignment via DHCP.



Network interface receiving IP parameters automatically from DHCP.

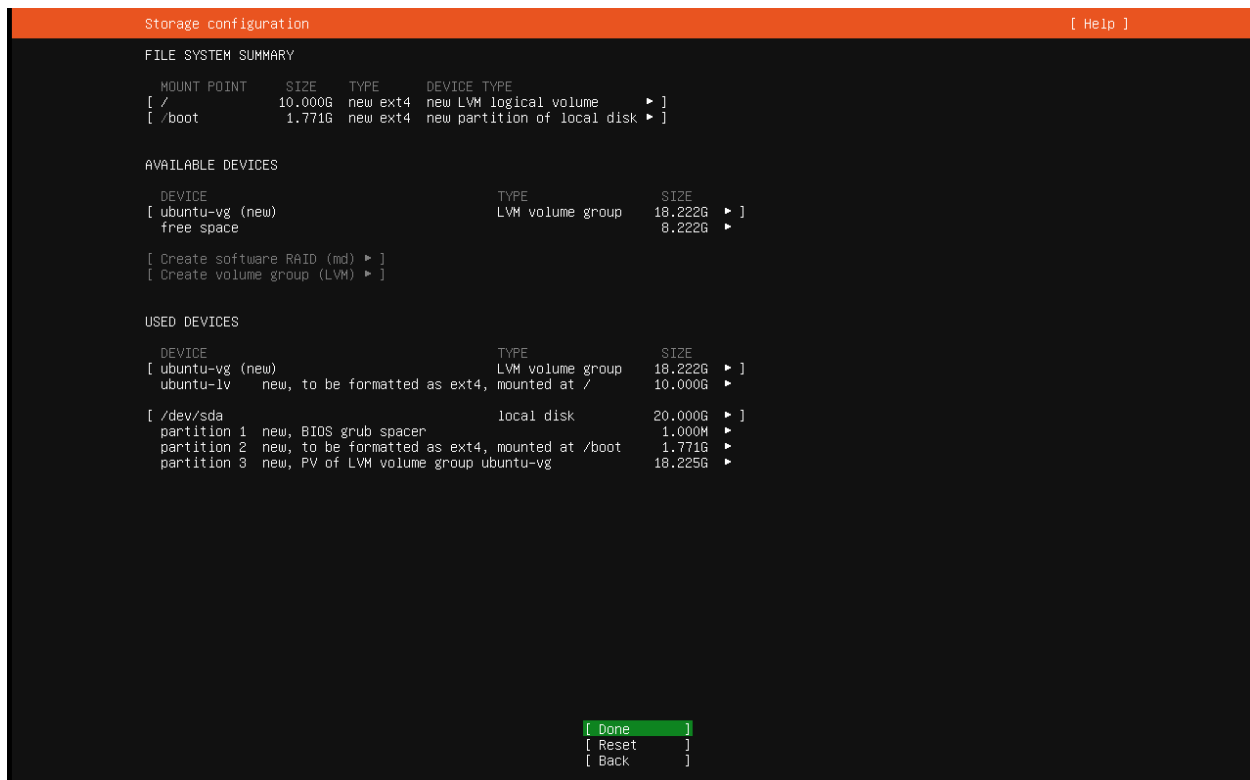
Task 9: Configure Storage

1. Selected guided storage allocation to target the dedicated VMware virtual disk.



Selecting guided disk partitioning options.

2. Reviewed layout parameters for the virtual disk before committing partition tables.

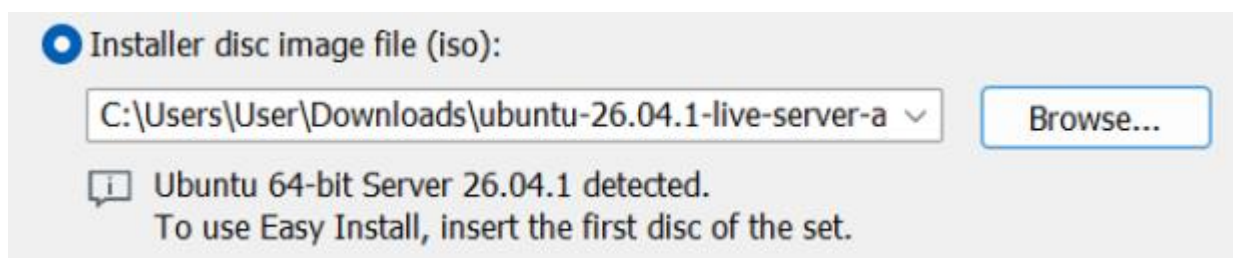


Storage layout summary screen before partition creation.

Task 10: Select the Installation Disk

1. Targeted the specific virtual disk provisioned within VMware.

```
[ /dev/sda local disk 20.000G ▼ ]
```



Virtual disk drive selection for system installation.

2. Cross-checked physical capacity and target identifier of the target drive.

```
AVAILABLE DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   18.222G ▶ ]
free space                             8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]
```

Detailed properties and size verification of the selected target disk.

Task 11: Review the Partitions

1. Evaluated proposed mount points, swap, and system partition structures.

```
USED DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   18.222G ▶ ]
ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ▶ ]

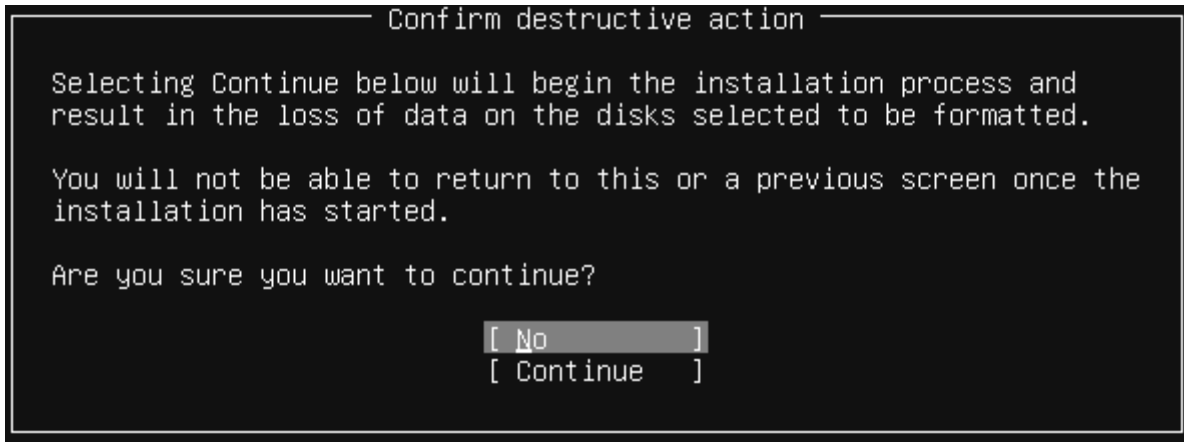
[ /dev/sda                             local disk                          20.000G ▶ ]
partition 1  new, BIOS grub spacer                      1.000M ▶
partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ▶
partition 3  new, PV of LVM volume group ubuntu-vg       18.225G ▶

[ Done ]
[ Reset ]
[ Back ]
```

Installer partition table layout view.

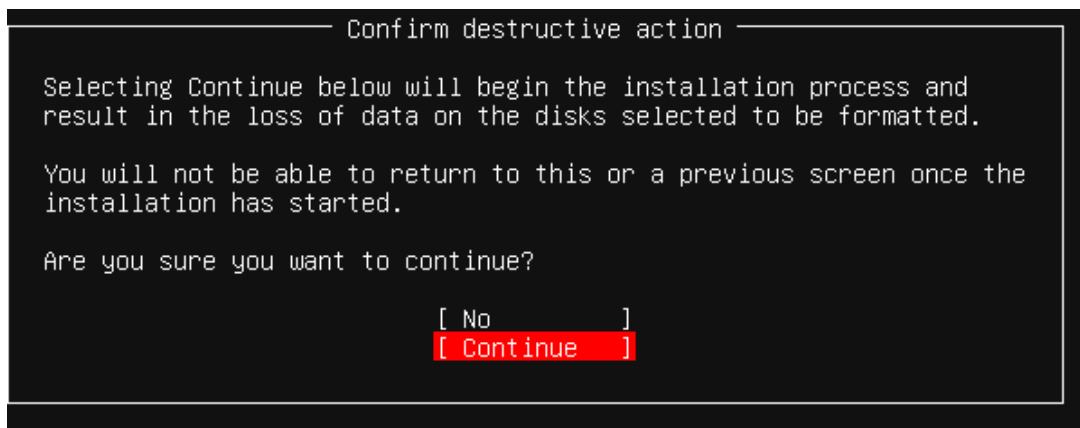
Task 12: Confirm the Storage Changes

1. Displayed warning prompt concerning destructive write actions on the target disk.



Storage destruction and commit warning prompt.

2. Confirmed and accepted disk partition rewrite operations.



Final confirmation step committing storage changes.

Task 13: Configure the Ubuntu Profile

1. Entered complete actual name in the profile configuration screen.



Profile configuration showing full legal name.

2. Set server hostname strictly to ubuntu.



Server hostname field set to ubuntu.

3. Set local account username identical to the GitHub username.



Pick a username: hifzashakeel152

Account username field showing exact GitHub username match.

4. Verified overall system profile data prior to installation execution.



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Hifza Shakeel

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: hifzashakeel152

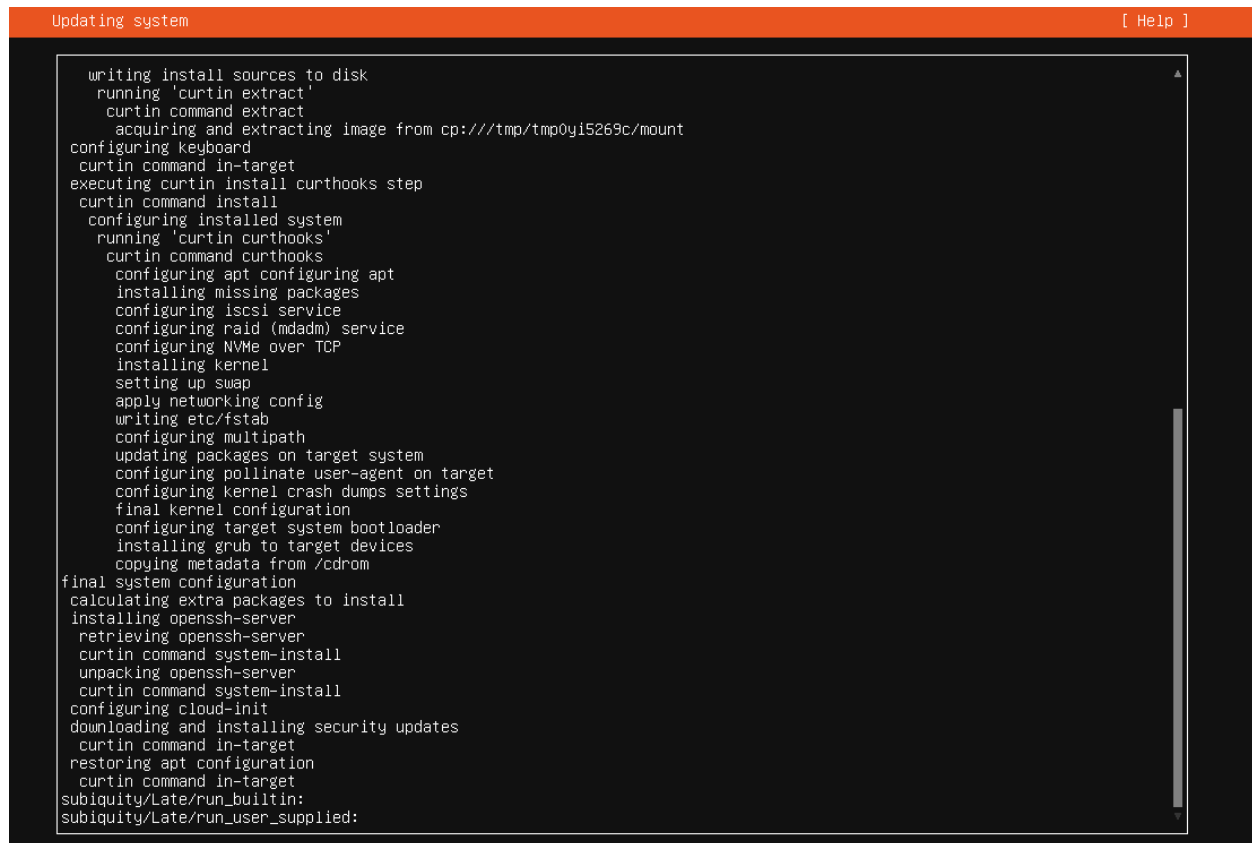
Choose a password: *****

Confirm your password: *****

Complete profile configuration summary prior to package deployment.

Task 14: Complete the Software Installation

1. Monitored installation process as base components and security packages deployed.



The screenshot shows a terminal window titled "Updating system" with a red header bar. The terminal displays a series of commands and their outputs, indicating the progress of the Ubuntu installer. The commands include writing install sources to disk, running 'curtin extract', curtin command extract, acquiring and extracting image from cp:///tmp/tmp0yi5269c/mount, configuring keyboard, curtin command in-target, executing curtin install curthooks step, curtin command install, configuring installed system, running 'curtin curthooks', curtin command curthooks, configuring apt, installing missing packages, configuring iscsi service, configuring raid (mdadm) service, configuring NVMe over TCP, installing kernel, setting up swap, apply networking config, writing etc/fstab, configuring multipath, updating packages on target system, configuring pollinate user-agent on target, configuring kernel crash dumps settings, final kernel configuration, configuring target system bootloader, installing grub to target devices, copying metadata from /cdrom, final system configuration, calculating extra packages to install, installing openssh-server, retrieving openssh-server, curtin command system-install, unpacking openssh-server, curtin command system-install, configuring cloud-init, downloading and installing security updates, curtin command in-target, restoring apt configuration, curtin command in-target, subiquity/Late/run_builtin:, and subiquity/Late/run_user_supplied:.

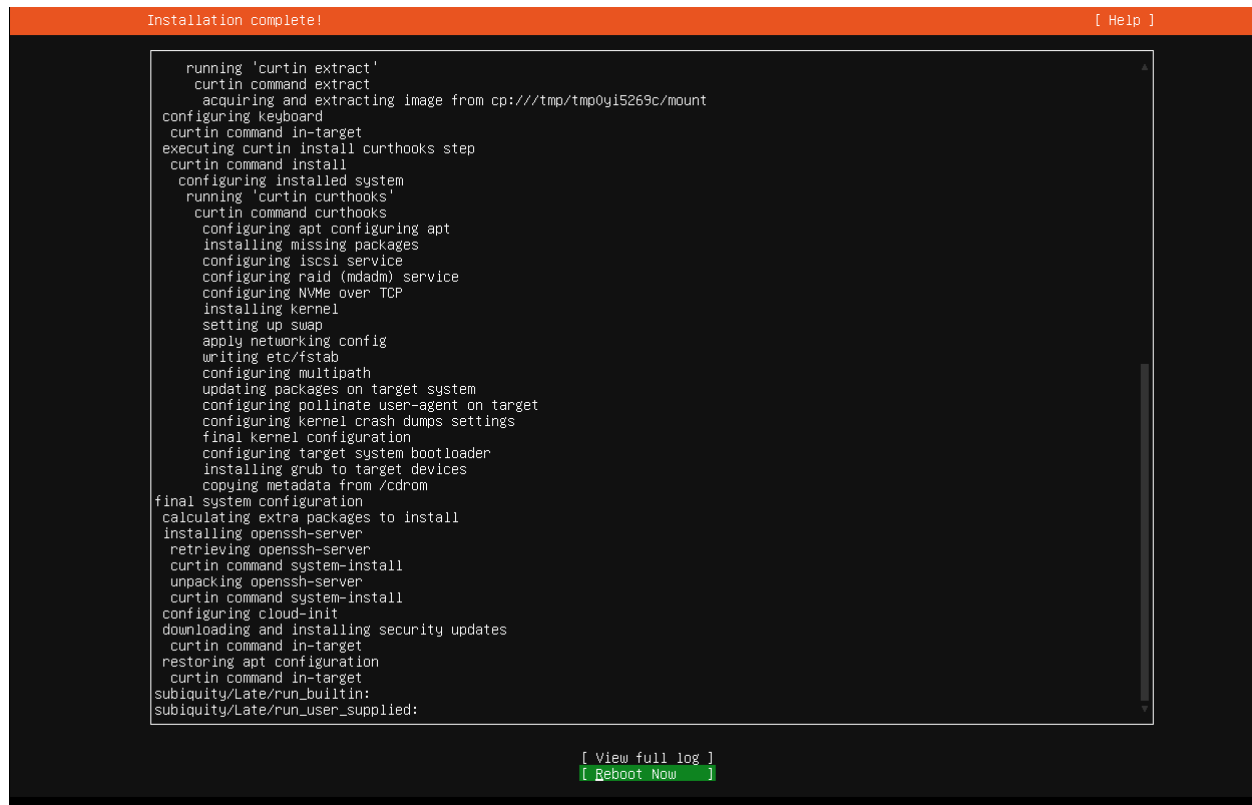
```
Updating system [ Help ]

writing install sources to disk
running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmp0yi5269c/mount
configuring keyboard
  curtin command in-target
executing curtin install curthooks step
  curtin command install
    configuring installed system
      running 'curtin curthooks'
        curtin command curthooks
          configuring apt
          installing missing packages
          configuring iscsi service
          configuring raid (mdadm) service
          configuring NVMe over TCP
          installing kernel
          setting up swap
          apply networking config
          writing etc/fstab
          configuring multipath
          updating packages on target system
          configuring pollinate user-agent on target
          configuring kernel crash dumps settings
          final kernel configuration
          configuring target system bootloader
          installing grub to target devices
          copying metadata from /cdrom
final system configuration
  calculating extra packages to install
  installing openssh-server
    retrieving openssh-server
    curtin command system-install
      unpacking openssh-server
      curtin command system-install
        configuring cloud-init
        downloading and installing security updates
        curtin command in-target
        restoring apt configuration
        curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

Ubuntu installer progress bar and package unpacking logs.

Task 15: Reboot and Verify the Ubuntu Identity

1. Installation completed successfully and requested system reboot.

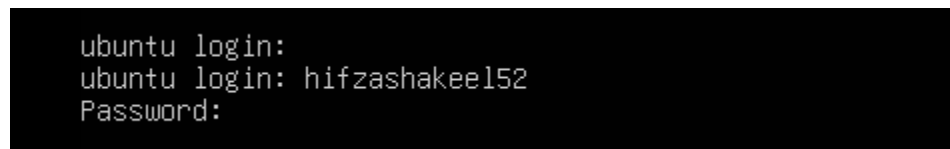


```
Installation complete! [ Help ]

running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/0y15269c/mount
configuring keyboard
  curtin command in-target
executing curtin install curthooks step
  curtin command install
    configuring installed system
      running 'curtin curthooks'
        curtin command curthooks
          configuring apt configuring apt
            installing missing packages
              configuring iscsi service
                configuring raid (mdadm) service
                  configuring NVMe over TCP
                    installing kernel
                      setting up swap
                        apply networking config
                          writing etc/fstab
                            configuring multipath
                              updating packages on target system
                                configuring pollinate user-agent on target
                                  configuring kernel crash dumps settings
                                    final kernel configuration
                                      configuring target system bootloader
                                        installing grub to target devices
                                          copying metadata from /cdrom
final system configuration
  calculating extra packages to install
    installing openssh-server
      retrieving openssh-server
        curtin command system-install
          unpacking openssh-server
            curtin command system-install
              configuring cloud-init
                downloading and installing security updates
                  curtin command in-target
                    restoring apt configuration
                      curtin command in-target
                        subiquity/Late/run_built-in:
                        subiquity/Late/run_user_supplied:
```

Installation complete message prompting system reboot.

2. System booted into installed Ubuntu instance presenting terminal authentication prompt.



```
ubuntu login:
ubuntu login: hifzashakeel52
Password:
```

Ubuntu Server TTY console login screen.

3. Logged in using the account credentials defined during setup.

```

ubuntu login:
ubuntu login: hifzashakeel52
Password:

Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 09:36:38 AM UTC 2026

System load:  0.0               Processes:            214
Usage of /:   50.5% of 9.75GB   Users logged in:     0
Memory usage: 21%              IPv4 address for ens3: 192.168.88.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

hifzashakeel52@ubuntu:~$
hifzashakeel52@ubuntu:~$ _

```

Successful console login showing terminal shell initialization.

4. Confirmed local identity prompt matches <github-username>@ubuntu:~\$ format.

```

hifzashakeel52@ubuntu:~$
hifzashakeel52@ubuntu:~$ whoami
hifzashakeel52
hifzashakeel52@ubuntu:~$ hostname
ubuntu
hifzashakeel52@ubuntu:~$ _

```

Shell prompt demonstrating strict adherence to identity formatting rules.

Task 16: Find the Ubuntu Server IP Address

1. Executed ip addr inside the local shell to list network interface details.

```

ubuntu
hifzashakeel52@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:b7:3e:1b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29b73e1b
    inet 192.168.88.130/24 metric 100 brd 192.168.88.255 scope global dynamic ens33
        valid_lft 1193sec preferred_lft 1193sec
    inet6 fe80::20c:29ff:feb7:3e1b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hifzashakeel52@ubuntu:~$ _

```

Terminal output following the execution of ip addr command.

2. Identified active network interface IPv4 address (excluding loopback 127.0.0.1).

```

        valid_lft 1193sec preferred_lft 1193sec
    inet6 fe80::20c:29ff:feb7:3e1b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hifzashakeel52@ubuntu:~$ _

```

Highlighting active network interface IPv4 address alongside identity prompt.

Task 17: Connect to Ubuntu Server from Windows

1. Initiated remote SSH connection from Windows terminal using command `ssh <github-username>@<ubuntu-server-ip>`

```

C:\Users\User>ssh hifzashakeel52@192.168.88.130
The authenticity of host '192.168.88.130 (192.168.88.130)' can't be established.
ED25519 key fingerprint is SHA256:l3rYeWxC8LE0kvNnmiuI75yF2mr5eHlOpPv570CUfaI.
This key is not known by any other names.

```

Executing SSH command from Windows host Command Prompt/PowerShell.

2. Accepted host key authenticity fingerprint verification on first connection.

```

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.88.130' (ED25519) to the list of known hosts.
hifzashakeel52@192.168.88.130's password:

```

Host fingerprint verification prompt and confirmation response.

3. Authenticated successfully over SSH connection using system account credentials.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.88.130' (ED25519) to the list of known hosts.
hifzashakeel52@192.168.88.130's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)
```

Established SSH session with banner output.

4. Verified remote shell prompt displays expected <github-username>@ubuntu format.

```
hifzashakeel52@ubuntu:~$ |
```

Remote shell window displaying correctly formatted identity prompt.

Task 18: Prepare Lab1 Solution PDF

1. Generated final output solution PDF file matching required naming and path specification.



File Explorer showing exported Lab1_Solution.pdf at Labs/Lab01/Lab1_Solution.pdf.